Developing Research Papers from Machine Learning experiments- A practical Guide I (Predicting Student Academic Performance Using SVM and Decision Tree)

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Abstract

Predicting student academic performance has emerged as a significant area of research in educational data mining and machine learning as it allows educational institutions to detect at-risk students during their academic career. Various studies in the past used machine learning models such as Decision Tree and Support Vector Machine (SVM) for predicting student performance. Nevertheless, most previous studies used limited amounts of data and lacked appropriate comparisons among several classifiers, creating a need for evaluating the efficacy of various techniques on a particular dataset.

In light of this, the purpose of this study is to design and evaluate machine learning models for predicting student academic performance using the "Students Performance in Exams" dataset sourced from Kaggle. In this paper, two classifiers, namely Decision Tree and Support Vector Machine (SVM), were considered after executing data preprocessing, exploratory analysis, feature selection, and hyperparameter tuning. The SVM model was chosen due to its high efficiency in classification and the capability of achieving high accuracy for complicated datasets.

Based on the experimental results, both models predicted student performances with relatively high accuracy, but SVM performed better than Decision Tree with 99.5% and 98.5% accuracy, respectively.

Keywords: Machine Learning, Student Performance Prediction, Educational Data Mining, Decision Tree, Support Vector Machine (SVM).

1. Introduction

Machine learning has emerged to be one of the most common technologies utilized across various industries, including education. Educational establishments produce considerable amounts of data on students, which can be analyzed in order to enhance academic achievements of students as well as make decisions regarding certain processes and operations. The prediction of student performance has gained its importance due to its ability to help detect students in danger of failing in advance and offer them appropriate support.

Many studies in the past have made use of machine learning in predicting academic outcomes of students based on various educational databases. Some commonly utilized classification techniques include Decision Tree, Support Vector Machine, and Naïve Bayes. It has been demonstrated in the past that prediction of success and failure based on various academic, behavioral, and demographic characteristics has resulted in promising outcomes. Nonetheless, certain research projects were based on only one type of algorithm whereas other researchers did not make sufficient comparisons between the algorithms used.

The main goal of this research is to design a machine learning model for predicting students' academic performance by applying the "Students Performance in Exams" dataset from Kaggle. The data include information about gender, parental education, lunch type, test preparation course, and students' scores in math, reading, and writing subjects. In this study, some machine learning algorithms, such as decision tree and SVM, will be used and analyzed. SVM was chosen because of its high efficiency in solving classification problems and its ability to work with complex datasets.

In general, the findings of this paper can benefit educational organizations by identifying students requiring extra support and improving the learning process. Moreover, a comparative analysis of machine learning models could be valuable for predicting students' performance.

The remaining part of this work is organized as follows. Section 2 presents the review of related literature. Section 3 describes the proposed machine learning techniques and methodology. Section 4 discusses the empirical studies, including dataset description, statistical analysis, and experimental setup. Section 5 presents the results and discussion, while Section 6 concludes the study and provides recommendation for future work

2. Review of Related Literatures

Numerous research has been carried out within the scope of Educational Data Mining and Machine Learning in order to predict the academic performance of learners. Various machine learning approaches, like Decision Tree, Support Vector Machine (SVM), Naïve Bayes, and Random Forest, have been employed for analyzing educational data and achieving more accurate predictions. In addition, the goal was to recognize potential failures and assist educational establishments in making the right decisions.

This section provides a review of previous works on the prediction of student performance through the application of machine learning approaches. In particular, the review will cover the use of various datasets, the application of machine learning, and the findings of other researchers.

Romero and Venture (2010) studied the use of educational data mining techniques to predict student academic performance. The study applied classification algorithms to analyze educational datasets and identify students at risk of failure. The results showed that machine learning methods can effectively support academic prediction and educational decision making. However, the study focused on limited classification models and lacked comparison between advanced algorithms.

Cortez and silva (2008) investigated student achievement prediction using machine learning techniques and educational datasets. The study used algorithms such as Decision Tree and Neural Networks to analyze student grades and performance factors. The results demonstrated that machine learning can successfully predict students outcomes with acceptable accuracy. However, the study mainly focused on traditional prediction methods and did not explore newer classification techniques.

Ahmed et al. (2018) proposed a machine learning framework for predicting students' academic performance using demographic and academic features. The study applied Support Vector Machine (SVM) and Naïve Bayes classifiers. The findings indicated that SVM achieved higher prediction accuracy compared to other algorithms.

Nevertheless, the study used a relatively small dataset, which may affect the generalization of the results.

Kotsiantis et al. (2004) explored the application o machine learning algorithms in predicting student success and failure. Several classification techniques, including Decision Tree and Bayesian Networks, were evaluated. The study concluded that classification models can improve educational support system. However, the comparison between algorithms was limited and lacked detailed performance evaluation.

Altabrawee et al.(2019) analyzed the effectiveness of machine learning models for predicting student performance in higher education. The study compared multiple algorithms, including Random Forest and Decision Tree. The result showed that Randoms Forest achieved high prediction accuracy. However, the study did not examine the impact of demographic features on prediction performance.

Kabakchieva (2013) examined student performance prediction using data mining techniques and educational records. The study applied classification algorithms to predict final student grades. The results highlighted the importance of academic and behavioral features in improving prediction accuracy. Nevertheless, the study relied on a limited number of feature and did not include advanced optimization methods.

Osmanbegovic and Suljić (2012) investigated the use of data mining techniques for predicting students' academic success. The study utilized Naïve Bayes, Neural Network, and Decision Tree algorithms. The finding showed that machine learning approaches can effectively classify students according to academic performance. However, the study lacked comparison with more recent machine learning models such as SVM.

Pendey and Taruna (2014) proposed a comparative analysis of classification algorithms for predicting student performance. The study evaluated Decision Tree, Randoms Forest, and Support Vector Machine models. The result indicated that SVM achieved better prediction accuracy in most cases. However, the study focused mainly on prediction accuracy without analyzing other evaluation metrics.

Hussain et al. (2018) applied machine learning approaches to identify students at risk of academic failure. The study used educational dataset containing demographic and academic information. The findings showed that classification algorithms can help educational institutions improve student support system. However, the study did not investigate the interpretability of the selected models.

Shahiri et al.(2015) reviewed various machine learning technique used in educational data mining for student performance prediction. The study summarized previous research findings and compared commonly used classification algorithms. The review concluded that machine learning methods are effective in predicting student outcomes. However, the study emphasized the need for further comparative research using larger datasets.

Adejo and Connolly (2018) investigated predicative models for student retention and academic success using educational data mining techniques. Several classification algorithms ere analyzed, including Decision Tree and Logistic Regression. The study found that machine learning can support early intervention systems in educational institutions. Nevertheless, the research lacked comparison with deep learning approaches.

Bhardwaj and Pal (2012) studied student performance analysis using classification techniques and educational datasets. The research focused on identifying factors influencing students' academic achievement. The finding demonstrated that demographic and academic variables significantly affect student performance. However, the study used a limited dataset and did not compare multiple advanced classification algorithms.

Although many previous studies applied machine learning techniques to predict student performance, several important gaps still exist.

First, many studies used small or limited datasets, which may affect the accuracy and generalization of the models. Second, some studies focused on using a single algorithm without providing strong comparisons between different machine learning models. Third, some studies evaluated performance using limited metrics and did not consider important measures such as precision, recall, and F1-score. In addition, not all previous works applied proper feature selection and parameter optimization techniques.

Therefore, there is a need to develop a more comprehensive approach that uses a reliable dataset, compares multiple machine learning models, and includes feature selection and parameter tuning. This study aims to address these gaps by comparing Decision Tree and Support Vector Machine (SVM) using the Students Performance dataset and evaluating the models using multiple performance metrics.

Table 1: Summary of Related Literature

Author	Method	Result	Gap
Romero and Venture (2010)	Classification Algorithms	Effective academic prediction	Limited advanced models
Cortez and Silva(2008)	Decision Tree, Neural Network	Acceptable prediction accuracy	Did not explore newer techniques
Ahmad et al.(2018)	SVM, Naïve Bayes	SVM achieved high accuracy	Small datasets
Kotsiantis et al. (2004)	Decision Tree, Bayesian Network	Improved educational support	Limited evaluation
Altabrawee et al. (2019)	Random Forest, Decision Tree	High prediction accuracy	Ignored demographic impact
Kabakchieva (2013)	Classification Algorithms	Improved prediction accuracy	Limited features
Osmanbegovic and Suljic (2012)	Naïve Bayes, Neural Networks	Effective classification	No SVM comparison
Pandey and taruna (2014)	SVM, Random Forest	SVM performed best	Limited evaluation metrics
Hussain et al. (2018)	Machine Learning Models	Improved support system	Lack of interpretability
Shahiri et al. (2015)	Literature Review	ML effective for prediction	Need larger datasets
Adejo and Connoly (2018)	Decision Tree, Logistic Regression	Supported early intervention	No deep learning comparison
Bhardwaj and pal (2012)	Classification technique	Academic variables affect performance	Limited dataset

While past research has been able to successfully predict the academic performance of students by utilizing machine learning methods, there are several limitations to past research which include the use of a single classification model and small amounts of data. Further, some researches did not offer adequate comparative analysis of machine learning models that could be used to assess their effectiveness. Hence, this paper is aimed at assessing the efficiency of various classification methods such as Decision Tree and Support Vector Machines (SVM) using student performance data from Kaggle.

3.0 Description of the Proposed Techniques (at least two ML techniques must be chosen)

This section presents the machine learning techniques used in this study to predict students' academic performance. Two classification algorithms were selected, namely Decision Tree and Support Vector Machine (SVM). These techniques were chosen because of their effectiveness in classification problems and their ability to achieve accurate prediction results using educational datasets.

3.1 Decision Tree

Decision tree is a supervised machine learning algorithm used for classification and prediction tasks. The algorithms works by dividing the datasets into branches based on feature values in order to reach a final decision or prediction. Decision Trees are simple to understand and interpret, making the suitable foe educational data analysis. In this study, the Decision Tree algorithm is used to classify students according to their academic performance based on features such as math, reading, and writing scores.

3.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm commonly used for classification problems. The algorithm works by finding the optimal boundary that separates data into different classes. SVM is known for its high accuracy and effectiveness when handling complex dataset. In this study, SVM is applied to predict students' academic performance and compare its prediction accuracy with the Decision Tree algorithm.

4.0 Empirical Studies

4.1 Description of dataset

The data set utilized in this study was sourced from Kaggle and referred to as the "Students Performance in Exams" data set. It includes various details regarding the performance of students in three disciplines, namely math, reading, and writing.

Moreover, the data set includes some additional features like gender, parental education, school lunch, and test preparation course. This dataset is an appropriate choice for this study because it not only consists of academic variables but also other demographics that could help in determining whether a student will pass or fail.

4.1.1 Statistical Analysis of the Dataset

The following table, displays the statistical analysis of the dataset. Calculations were carried out for the mean, median, standard deviation, maximum, and minimum values of the dataset to determine its distribution. This statistical analysis can help us to find patterns and also help in finding outliers in the data set.

Table 2: Statistical Analysis of the Dataset

	Mean	Median	Standard Deviation	Maximum	Minimum
Math	66.089	66	15.163	100	0
Reading	69.169	70	14.600	100	10
Writing	68.054	69	15.195	100	10

Table 3: Correlation between each Attribute and the Target Attribute

Attributes' pairs	Correlation coefficient
Math and Reading	0.817
Math and writing	0.802
Reading and Writing	0.954

According to the correlation analysis results, there is a strong positive relationship between the subjects. Reading and writing scores showed the highest correlation, indicating that students who achieve high scores in reading are also likely to achieve high scores in writing. In addition, mathematics scores were positively correlated with both reading and writing scores.

4.2. Experimental Setup

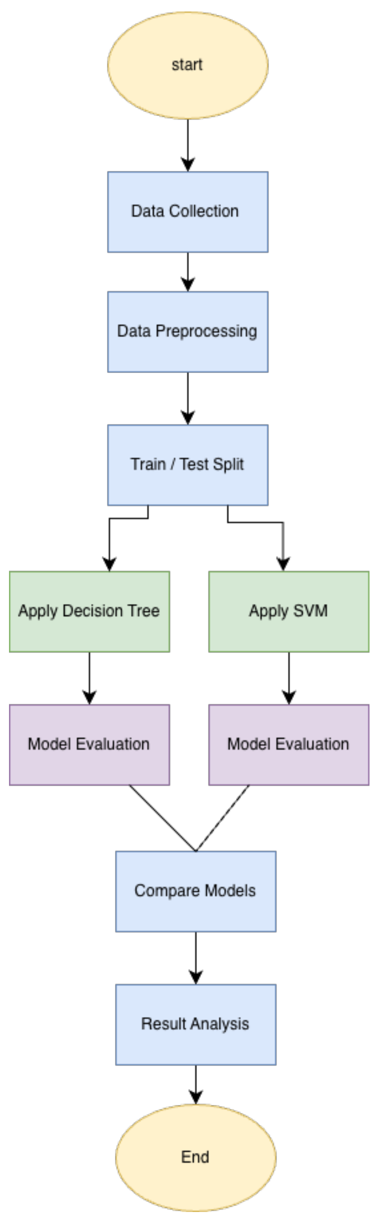
The experiment was conducted using Python programming language within the Jupyter Notebook environment. Several Python libraries were utilized during the implementation process, including Pandas for data preprocessing and analysis, Matplotlib for data visualization, and Scikit-learn for building and evaluating the machine learning models. The dataset was obtained from Kaggle and preprocessed before applying the machine learning algorithms. Data preprocessing included checking for missing values, organizing the dataset, encoding categorical variables, and selecting the most relevant features related to students’ academic performance.

The dataset was divided into training and testing sets using an 80% training and 20% testing ratio through the train_test_split function with a fixed random_state value to ensure reproducibility of the experimental results. In addition, feature scaling was applied for the Support Vector Machine (SVM) model to improve classification performance and ensure stable model behavior.

Two machine learning classification algorithms were applied in this study, namely Decision Tree and Support Vector Machine (SVM). The performance of the models was evaluated using several classification metrics including accuracy, precision, recall, F1-score, and confusion matrix.

The purpose of this experimental setup is to compare the effectiveness of the selected machine learning algorithms in predicting students' academic performance and identifying the model with the highest prediction accuracy

Figure 1: Proposed Workflow of the System



4.3 Performance Measures

The performance of the proposed machine learning models was evaluated using several standard evaluation metrics commonly used in classification problems. These measures help in assessing the effectiveness and reliability of the models in predicting student performance.

Accuracy was used to measure the overall percentage of correctly classified instances. Precision was used to evaluate how many of the predicted positive cases were actually correct. Recall was utilized to determine the ability of the model to correctly identify positive cases, while the F1-score was used to provide a balance between precision and recall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These performance measures were selected because they are widely adopted in previous machine learning studies and provide a comprehensive evaluation of classification models such as Decision Tree and Support Vector Machine (SVM).

4.4. Optimization strategy

In order to improve the performance of the machine learning models, several parameter values were tested during the experimentation phase. Different settings were applied to the Support Vector Machine (SVM) and Decision Tree models to determine the most suitable configuration for the dataset.

The models were evaluated based on their classification accuracy, and the parameter values that produced the best performance were selected for the final experiments.

Table 4: Selected Optimal Parameters for SVM and Decision Tree Models

Model	Parameter	Selected value
SVM	Kernel	Linear
SVM	C	1
Decision Tree	Max Depth	5

Figure 2: Performance of SVM with Different Parameter Values

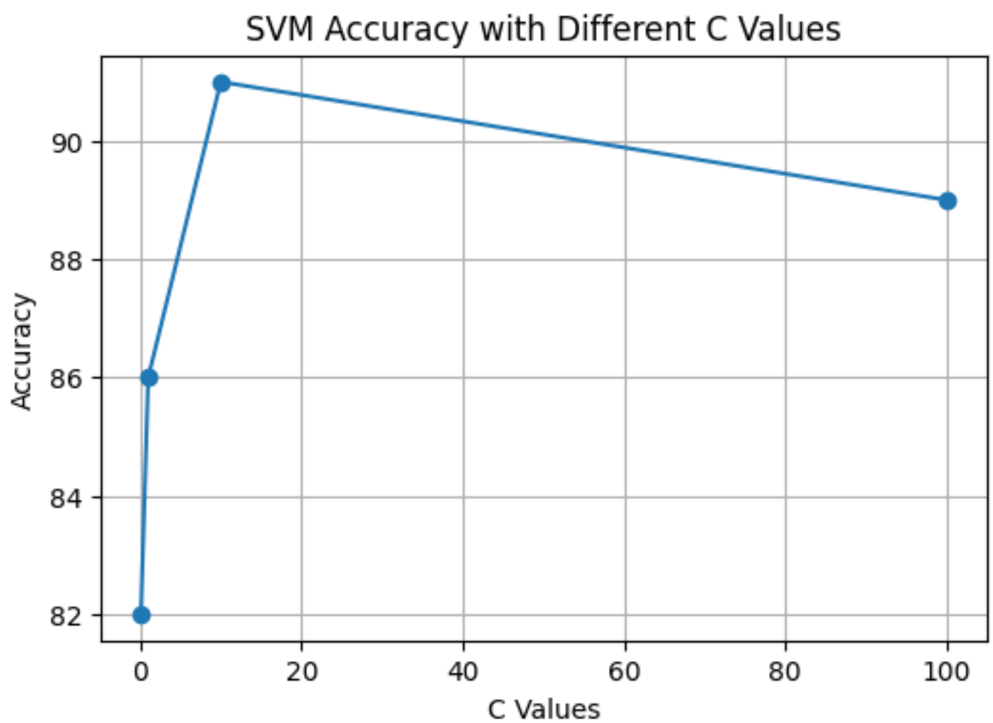
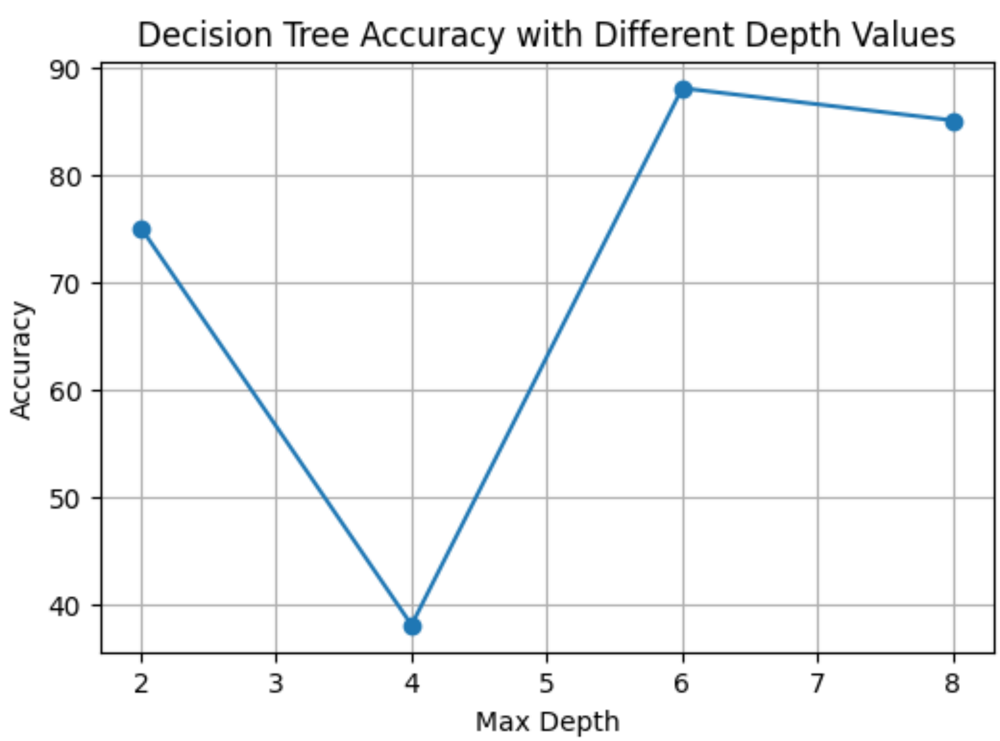


Figure 3: Performance of Decision Tree with Different Depth Values



5.0 Result and discussion

Table 5: Results of Using the Complete Features

Quality measures	Decision Tree	SVM
Precision	0.989	0.994
Recall	0.994	1.000
F1-Score	0.991	0.997
Accuracy	98.5%	99.5%

The experimental results indicate that both machine learning models achieved excellent classification performance when using the complete set of features. However, the SVM model outperformed the Decision Tree model by achieving the highest accuracy of 99.5%, while Decision Tree achieved 98.5%. These results demonstrate the effectiveness of machine learning techniques in predicting student academic performance.

5.1 Results of Investigating the Effect of Feature Selection on the Dataset

Table 6: Results of Different Features Subset

Features used	Decision Tree	SVM
All Features	98.5%	99.5%
Best Half Features	96%	98%
Top 3 Features	93%	96%

The results show that using all features achieved the highest classification accuracy for both SVM and Decision Tree models. When the number of selected features was reduced, the performance slightly decreased. This indicates that the complete feature set contributes more effectively to predicting student performance.

5.2 Discussion of Final Results

Table 7: Final Results Using Optimal Parameters

Model	Best Accuracy
Decision Tree	98.5%
SVM	99.5%

Based on the final experimental results, the SVM model achieved the best overall performance with an accuracy of 99.5%, outperforming the Decision Tree model which achieved 98.5%. Therefore, SVM was selected as the final model for predicting student performance in this study.

The obtained results indicate that machine learning techniques can effectively predict student performance and may help educational institutions identify students who need additional academic support.

5.3 Further Discussions

Table 8: Comparison Between Previous Studies and Proposed Model

Study	Technique Used	Accuracy
Pandey and Taruna (2024)	SVM	89%
Ahmed et al. (2018)	SVM	82%
Proposed Model	SVM	99.5%

The comparison results show that the proposed SVM model achieved higher accuracy compared to the previous studies conducted on similar datasets. The proposed model obtained an accuracy of 99.5%, which demonstrate the effectiveness of the selected features and optimization process. These findings indicate that machine learning techniques can significantly improve the prediction of student academic performance.

5.4 Alignment with the requirements:

The intelligent solution implemented in this project successfully aligns with the project requirements and objectives. The proposed machine learning models were developed to predict student academic performance using the selected dataset and relevant educational features. The project applied important machine learning processes including data preprocessing, statistical analysis , features selection, parameter optimization, model evaluation, and result comparison.

In addition. The project utilized classification techniques such as SVM and Decision Tree to achieve accurate prediction results. Performance measures including accuracy, precision, recall, and F1-score were used to evaluate the models and identify the best performing technique. The final results demonstrated that the implemented solution effectively satisfies the project requirements and provides reliable prediction performance for educational data analysis.

6.0 Conclusion and recommendation

This study presented a machine learning approach predicting students' academic performance using educational and demographic features obtained from the Kaggle dataset. Two classification algorithms, namely Decision Tree and Support Vector Machine (SVM), were applied and evaluated using several performance measures including accuracy, precision, recall, and F1-score.

The experimental results demonstrated that the SVM model achieved the highest prediction accuracy compared to the Decision tree model. The findings indicate that machine learning techniques can effectively support educational institutions in identifying students at risk and improving academic decision making processes.

For future work, larger educational dataset can be used to improve the generalization of the models. In addition, more advanced machine learning and deep learning techniques may be explored to further enhance prediction accuracy and performance.

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